

## 1. Find the Largest Element in an Array

### Problem Statement:

Given an array of integers, find and return the largest element.

### Example:

- **Input:** [5, 2, 9, 1, 5]
- **Output:** 9

## 2. Find the Smallest Element in an Array

### Problem Statement:

Given an array of integers, find and return the smallest element.

### Example:

- **Input:** [5, 2, 9, 1, 5]
- **Output:** 1

## 3. Reverse an Array

### Problem Statement:

Given an array, reverse its elements **in-place** (without using extra space).

### Example:

- **Input:** [1, 2, 3, 4, 5]
- **Output:** [5, 4, 3, 2, 1]

## 4. Check if an Array is Sorted

### Problem Statement:

Given an array, check if it is sorted in **non-decreasing order**. Return true if sorted, else false.

### Example:

- **Input:** [1, 2, 3, 4, 5] → **Output:** true
- **Input:** [5, 2, 7, 1] → **Output:** false

## 5. Remove Duplicates from Sorted Array

### Problem Statement:

Given a **sorted** array of integers, remove duplicates **in-place** and return the new length.

### Example:

- **Input:** [1, 1, 2, 2, 3, 4, 4]
- **Output:** 4 (Modified array: [1, 2, 3, 4, ...])

## 6. Move All Zeroes to the End

### Problem Statement:

Given an array, move all 0s to the end while maintaining the order of non-zero elements.

### Example:

- **Input:** [0, 1, 0, 3, 12]
- **Output:** [1, 3, 12, 0, 0]

## 7. Find the Intersection of Two Arrays

### Problem Statement:

Given two integer arrays, return an array of their common elements (intersection).

### Example:

- **Input:** nums1 = [1, 2, 2, 1], nums2 = [2, 2]
- **Output:** [2]

## 8. Rotate Array by One Position

### Problem Statement:

Given an array, rotate it **clockwise by one** position (shift all elements right by one, last element moves to first).

### Example:

- **Input:** [1, 2, 3, 4, 5]
- **Output:** [5, 1, 2, 3, 4]

## 9. Find the Missing Number (1 to n)

### Problem Statement:

Given an array of size  $n-1$  containing distinct integers from 1 to  $n$ , find the missing number.

### Example:

- **Input:** [1, 2, 4, 5] ( $n=5$ )
- **Output:** 3

## 10. Find the Number Occurring Odd Number of Times

### Problem Statement:

Given an array where all numbers occur **even** times except one, find that number.

### Example:

- **Input:** [4, 3, 4, 4, 4, 2, 2]
- **Output:** 3

## 11. Find the First Repeating Element

### Problem Statement:

Given an array, find the first element that appears more than once. Return -1 if no duplicates exist.

### Example:

- **Input:** [3, 2, 1, 2, 3]
- **Output:** 2